

KEWEI WANG

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EDUCATION

Courant Institute of Mathematical Sciences, New York University

New York, NY

Ph.D. in Mathematics (expected), advised by Prof. Georg Stadler

2024 - 2029

Peking University

Beijing, China

B.S. in Computational Mathematics

2020 - 2024

RESEARCH EXPERIENCE

Research interests: optimization under uncertainty, PDE-constrained optimization, Bayesian inverse problems, and relevant applications.

Chance constrained optimization (current research)

Dec. 2024 - Present

Advised by Prof. Georg Stadler, NYU Courant

- Working on chance constrained optimization and its application to Gaussian process regression.

Inverse problems in multipopulation mean field games

Apr. 2023 - Feb. 2024

Advised by Prof. Kui Ren, Columbia University

- Generalized the model of single population MFG to multipopulation cases under several frameworks. Developed and proved several results about uniquely reconstructing cost functions, one of which shows the possibility of unique reconstruction of cost functions from data measured only on a single population component.

PUBLICATION

- [1] K. Ren, N. Soedjak, and K. Wang, "Unique determination of cost functions in a multipopulation mean field game model," *Journal of Differential Equations*, vol. 427, pp. 843–867, 2025.
- [2] K. Ren, N. Soedjak, K. Wang, and H. Zhai, "Reconstructing a state-independent cost function in a mean-field game model," *Inverse Problems*, vol. 40, no. 10, p. 105 010, 2024.

TEACHING

Teaching Assistant at NYU Courant

- Math for Economics II

Fall 2025

AWARDS

- Henry M. MacCracken Fellowship
Full Ph.D. support at NYU 2024-2029
- Silver Award, Lin Award (Applied and Computational Mathematics) in S.-T. Yau College Student Mathematics Contest
June 2023

MISCELLANEOUS

- Programming languages: MATLAB, Python.
- Languages: English - Fluent, Chinese - Native.